

JADAVPUR UNIVERSITY
B.E. ELECTRICAL ENGINEERING EXAMINATION - 2023
ELECTRICAL MACHINES - II (Part I)

Detailed Analytical Solutions provided by SR Sir

Question 1(a): What is the speed of rotor mmf of a 3-phase induction motor with respect to its stator mmf? - Justify your answer.

SR Sir's Thought Process

This is a fundamental kinematic analysis of magnetic fields in AC machines. The stator produces an MMF rotating at N_s in space. The rotor currents are at slip frequency (sf_1), so the rotor MMF rotates at sN_s relative to the rotor. The rotor itself rotates at $(1-s)N_s$. Adding them up shows the rotor MMF rotates at N_s in space. Thus, relative speed is zero.

Answer:

The speed of the rotor MMF with respect to the stator MMF in a 3-phase induction motor is **Zero**.

Justification:

- When a balanced 3-phase supply is given to the stator, it produces a rotating magnetic field (stator MMF) that rotates at synchronous speed $N_s = \frac{120f_1}{P}$ with respect to the stationary stator (in space).
- Let the rotor rotate at a mechanical speed of N_r . The slip s is defined as $s = \frac{N_s - N_r}{N_s}$, which means $N_r = (1-s)N_s$.
- The relative speed of the stator MMF with respect to the rotor conductors is $N_s - N_r = sN_s$. This relative motion induces currents in the rotor at slip frequency $f_2 = sf_1$.
- These slip-frequency rotor currents produce their own rotating magnetic field (rotor MMF). The speed of this rotor MMF *with respect to the rotor structure* is $N_{\text{rotor-field}} = \frac{120f_2}{P} = \frac{120(sf_1)}{P} = sN_s$.
- To find the absolute speed of the rotor MMF in space (i.e., with respect to the stationary stator), we must add the mechanical speed of the rotor body to the speed of the rotor MMF relative to the rotor:

$$\text{Speed of rotor MMF in space} = N_r + sN_s = (1-s)N_s + sN_s = N_s \quad (1)$$

Since both the stator MMF and the rotor MMF rotate at the exact same synchronous speed (N_s) in the same direction, they are magnetically locked. The relative speed between them is exactly zero.

Question 1(b): Why stators in a 3-phase induction motor (except large rating ones) are usually provided with semi-closed type slots? Explain.

SR Sir's Thought Process

Slot design is a trade-off between electrical/magnetic performance and manufacturing ease. Open slots make coil insertion easy (pre-formed coils), but increase the effective air gap, leading to high magnetizing current and poor power factor. Semi-closed slots reduce the air gap reluctance while still allowing conductor wires to be fed through the narrow opening. I will list the pros and cons clearly.

Answer:

Stators of small to medium rating 3-phase induction motors are predominantly provided with **semi-closed type slots** due to a highly beneficial compromise between electromagnetic performance and manufacturing constraints.

Explanation / Advantages of Semi-Closed Slots:

1. **Reduced Air-Gap Reluctance:** The narrow opening of a semi-closed slot ensures that the effective area of the magnetic flux path across the air gap is larger compared to open slots. A lower magnetic reluctance means less magnetizing current (I_m) is required to establish the necessary air-gap flux.
2. **Improved Power Factor:** Because the magnetizing current I_m (which is highly inductive and lags the voltage by nearly 90°) is reduced, the overall no-load and full-load operating power factor of the motor is significantly improved.
3. **Reduced Tooth Pulsation Losses:** A smoother inner stator periphery minimizes the high-frequency variations in magnetic flux density on the rotor surface (slot harmonics), thereby reducing surface iron losses (pulsation losses) and minimizing magnetic acoustic noise.
4. **Higher Average Flux Density:** It allows for a more uniform flux distribution, leading to better utilization of the magnetic core material.

Manufacturing constraint: The only drawback is that pre-formed solid coils cannot be dropped directly into semi-closed slots. Individual turns must be fed through the narrow slit one by one, making manufacturing slightly more labor-intensive, which is why very large machines (which use massive rigid copper bars) resort to open slots.

Question 1(c): An 8-pole, 3-phase, 60° spread, double layer winding has 72 coils in 72 slots. The coils are short-pitched by two slots. Calculate the winding factor for the fundamentals and third harmonics.

SR Sir's Thought Process

This is a direct numerical application of pitch factor (K_p) and distribution factor (K_d). Total slots $S = 72$, Poles $P = 8$. Slots per pole $= 72/8 = 9$. Slot angle $\gamma = 180^\circ/9 = 20^\circ$. Slots per pole per phase $m = 72/(8 \times 3) = 3$. Short pitched by 2 slots $\Rightarrow$ chording angle $\alpha = 2 \times 20^\circ = 40^\circ$. $K_{pn} = \cos(n\alpha/2)$. $K_{dn} = \frac{\sin(nm\gamma/2)}{m \sin(n\gamma/2)}$. Winding factor $K_{wn} = K_{pn} \times K_{dn}$. Need to calculate for $n = 1$ and $n = 3$.

Answer:

Given Data:

- Number of poles (P) = 8
- Number of phases = 3
- Phase spread = 60°
- Total slots (S) = 72
- Short pitching = 2 slots

Step 1: Determine winding parameters Slots per pole (S_p) = $\frac{S}{P} = \frac{72}{8} = 9$ slots/pole. Slot electrical angle (γ) = $\frac{180^\circ}{S_p} = \frac{180^\circ}{9} = 20^\circ$. Slots per pole per phase (m) = $\frac{S}{P \times 3} = \frac{72}{24} = 3$. Chording angle (α) due to short-pitching by 2 slots = $2 \times \gamma = 2 \times 20^\circ = 40^\circ$.

Step 2: Calculate for Fundamental ($n = 1$) Pitch factor (K_{p1}):

$$K_{p1} = \cos\left(\frac{\alpha}{2}\right) = \cos\left(\frac{40^\circ}{2}\right) = \cos(20^\circ) = 0.9397 \quad (2)$$

Distribution factor (K_{d1}):

$$K_{d1} = \frac{\sin\left(m\frac{\gamma}{2}\right)}{m \sin\left(\frac{\gamma}{2}\right)} = \frac{\sin\left(3 \times \frac{20^\circ}{2}\right)}{3 \sin\left(\frac{20^\circ}{2}\right)} = \frac{\sin(30^\circ)}{3 \sin(10^\circ)} = \frac{0.5}{3 \times 0.1736} = 0.9598 \quad (3)$$

Fundamental Winding Factor (K_{w1}):

$$K_{w1} = K_{p1} \times K_{d1} = 0.9397 \times 0.9598 = \mathbf{0.9019} \quad (4)$$

Step 3: Calculate for Third Harmonic ($n = 3$) Pitch factor (K_{p3}):

$$K_{p3} = \cos\left(\frac{n\alpha}{2}\right) = \cos\left(\frac{3 \times 40^\circ}{2}\right) = \cos(60^\circ) = 0.5000 \quad (5)$$

Distribution factor (K_{d3}):

$$K_{d3} = \frac{\sin\left(nm\frac{\gamma}{2}\right)}{m \sin\left(n\frac{\gamma}{2}\right)} = \frac{\sin\left(3 \times 3 \times \frac{20^\circ}{2}\right)}{3 \sin\left(3 \times \frac{20^\circ}{2}\right)} = \frac{\sin(90^\circ)}{3 \sin(30^\circ)} = \frac{1.0}{3 \times 0.5} = 0.6667 \quad (6)$$

Third Harmonic Winding Factor (K_{w3}):

$$K_{w3} = K_{p3} \times K_{d3} = 0.5000 \times 0.6667 = \mathbf{0.3333} \quad (7)$$

Question 1(d): What is the relationship of developed torque of a 3-phase induction motor with the supply voltage? Justify your answer.

SR Sir's Thought Process

Torque depends on flux (Φ) and rotor current (I_2). Both of these quantities are directly proportional to the applied voltage V_1 . Thus, $T \propto V_1^2$. I will show this strictly through the torque equation to leave no room for ambiguity.

Answer:

The electromagnetic torque (T_e) developed by a 3-phase induction motor is **directly proportional to the square of the supply voltage** (V_1^2).

$$T_e \propto V_1^2 \quad (8)$$

Justification: From the equivalent circuit of an induction motor, the general expression for the developed electromagnetic torque is given by:

$$T_e = \frac{3}{\omega_s} \frac{V_1^2 \left(\frac{R'_2}{s} \right)}{\left(R_1 + \frac{R'_2}{s} \right)^2 + (X_1 + X'_2)^2} \quad (9)$$

In this equation, for a given constant slip s and fixed motor parameters (R_1, X_1, R'_2, X'_2), the entire denominator and the constant multipliers act as a single proportionality constant (k). Therefore:

$$T_e = kV_1^2 \quad (10)$$

Physical interpretation: Torque is produced by the interaction of the air-gap flux (Φ) and the active component of the rotor current ($I_2 \cos \theta_2$). 1. The air-gap flux is directly proportional to the supply voltage ($\Phi \propto V_1$). 2. The induced rotor EMF E_2 is proportional to Φ , and thus the rotor current I_2 is also directly proportional to the supply voltage ($I_2 \propto V_1$). Since $T_e \propto \Phi \cdot I_2$, multiplying these two linear proportionalities gives $T_e \propto V_1 \cdot V_1 = V_1^2$. If the supply voltage drops by 10%, the torque drops by nearly 19%.

Question 1(e): Show that a 3-phase winding carrying sinusoidal currents produces space harmonics of the order of $h = 6k \pm 1$, where k is a positive integer.

SR Sir's Thought Process

The MMF waveform of a single phase is a stepped/rectangular spatial wave. Fourier series analysis shows it contains fundamental and all odd harmonics (due to half-wave symmetry). Even harmonics are absent. Triplen harmonics (3, 9, 15) are in space phase but differ in time phase by $120^\circ \times 3 = 360^\circ$, making them co-phasal. Since neutral is isolated, they cancel out in a 3-phase system. What remains are odd, non-triplen harmonics: 1, 5, 7, 11, 13... which can be written algebraically as $6k \pm 1$.

Answer:

When a 3-phase winding is fed with balanced sinusoidal currents, the resulting spatial Magnetomotive Force (MMF) distribution is not perfectly sinusoidal due to the discrete distribution of coils in slots. The spatial MMF wave takes the form of a stepped wave.

Proof of $h = 6k \pm 1$ Harmonics: 1. **Half-Wave Symmetry:** Because the north and south poles generated by the winding are physically identical but opposite in magnetic polarity, the spatial MMF wave possesses half-wave symmetry: $F(\theta) = -F(\theta + \pi)$. According to Fourier analysis, any waveform with half-wave symmetry contains **only odd harmonics**. Therefore, the harmonic order h can only be: $h = 1, 3, 5, 7, 9, 11, 13, 15 \dots$

2. **Cancellation of Triplen Harmonics in 3-Phase Systems:** Let us analyze the 3rd harmonic (and its multiples, the triplen harmonics: 3, 9, 15...). The spatial displacement between phase A and phase B for the fundamental wave is 120° . For the h -th spatial harmonic, this displacement becomes $h \times 120^\circ$. For the 3rd harmonic ($h = 3$), the phase displacement is: $3 \times 120^\circ = 360^\circ \equiv 0^\circ$ (electrical). This means the 3rd harmonic spatial MMFs of all three phases are perfectly in phase with each other (co-phasal) in space. Since the input currents form a balanced 3-phase system without a neutral return wire ($i_A + i_B + i_C = 0$), the algebraic sum of their co-phasal 3rd harmonic MMFs at any instant is zero. Therefore, all **triplen harmonics (multiples of 3) are completely eliminated** from the resultant air-gap flux.

3. **Resultant Harmonics:** Taking the set of odd numbers and removing the multiples of 3, the remaining harmonic orders are: $h = 1, 5, 7, 11, 13, 17, 19 \dots$. This exact mathematical sequence can be elegantly expressed by the formula:

$$\boxed{h = 6k \pm 1} \quad (11)$$

Where k is a positive integer ($k = 1, 2, 3 \dots$). For $k = 1$: $h = 6(1) - 1 = 5$ and $h = 6(1) + 1 = 7$. For $k = 2$: $h = 6(2) - 1 = 11$ and $h = 6(2) + 1 = 13$, and so forth.

Question 1(f): Show that, in an three phase induction motor, the 5^{th} harmonic flux produces backward rotating torque at a speed of $1/5^{th}$ synchronous speed whereas 7^{th} harmonic produces forward rotating torque at a speed of $1/7^{th}$ synchronous speed.

SR Sir's Thought Process

This is an extension of the previous question. We look at the phase sequence of the space harmonics. Fundamental phase difference: 120° . For the h -th harmonic, phase angle becomes $h \times 120^\circ$. For $h = 5$: $5 \times 120^\circ = 600^\circ$. $600^\circ - 360^\circ = 240^\circ \equiv -120^\circ$. This creates an ACB phase sequence (reverse). Speed is $N_s/5$. For $h = 7$: $7 \times 120^\circ = 840^\circ$. $840^\circ - 2 \times 360^\circ = 120^\circ$. This creates an ABC phase sequence (forward). Speed is $N_s/7$.

Answer:

The behavior of spatial harmonics depends on their effective phase sequence, which dictates the direction of rotation, and their pole count, which dictates their synchronous speed.

1. Speed of Rotation: A space harmonic of order h behaves exactly like a machine with $h \times P$ poles, where P is the number of fundamental poles. Since synchronous speed is given by $N_s = \frac{120f}{P}$, the synchronous speed of the h -th harmonic field is:

$$N_{s,h} = \frac{120f}{hP} = \frac{1}{h}N_s \quad (12)$$

Thus, the 5th harmonic rotates at $N_s/5$, and the 7th harmonic rotates at $N_s/7$.

2. Direction of Rotation (Phase Sequence Analysis): The phase sequence of the fundamental supply currents is A-B-C, meaning Phase B lags Phase A by 120° , and Phase C lags Phase A by 240° . For the h -th harmonic, the electrical phase displacement between the phase MMFs is multiplied by h .

Analysis of the 5th Harmonic ($h = 5$):

- Phase displacement for 5th harmonic = $5 \times 120^\circ = 600^\circ$.
- Subtracting 360° to find the principal angle: $600^\circ - 360^\circ = 240^\circ$ (or -120°).
- This means Phase B now lags Phase A by 240° , which is equivalent to Phase B *leading* Phase A by 120° .
- The effective phase sequence is reversed to **A-C-B**.
- Because the phase sequence is reversed, the 5th harmonic rotating magnetic field rotates in the **backward direction** (opposite to the fundamental). It induces a braking torque at a synchronous speed of $-N_s/5$.

Analysis of the 7th Harmonic ($h = 7$):

- Phase displacement for 7th harmonic = $7 \times 120^\circ = 840^\circ$.
- Subtracting multiples of 360° : $840^\circ - 720^\circ = 120^\circ$.
- Phase B lags Phase A by exactly 120° , identical to the fundamental.
- The effective phase sequence remains **A-B-C**.
- Because the phase sequence is the same, the 7th harmonic rotating magnetic field rotates in the **forward direction**. It induces a motoring torque at a synchronous speed of $+N_s/7$ (which causes the "crawling" phenomenon).

Question 1(g): What is deep bar rotor induction motor? How it produces high starting torque?

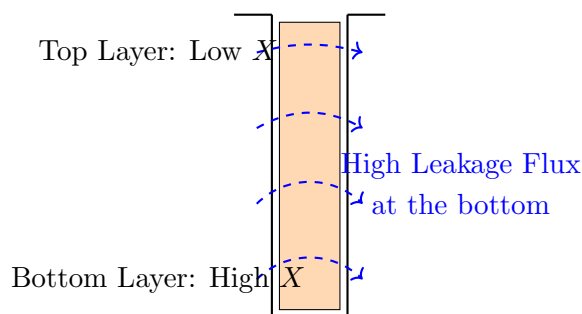
SR Sir's Thought Process

This question is identical to 1(f) from the 2022 paper. I will replicate the core physical principles regarding the Skin Effect, slot leakage flux, and how effective R_2 changes dynamically with slip frequency.

Answer:

Deep Bar Rotor Induction Motor: A deep bar rotor is a special construction of a squirrel cage rotor designed to inherently provide high starting torque without sacrificing the efficiency of low-slip running conditions. Instead of using standard circular or shallow square conductors, the rotor slots are punched very deep and narrow, and the copper/aluminum bars inserted into these slots possess a high depth-to-width ratio.

Mechanism of Producing High Starting Torque (Skin Effect): The operation relies entirely on the **Skin Effect**, which is governed by the frequency of the rotor currents. If we conceptually divide a deep rotor bar into multiple horizontal layers, the bottom-most layer (deepest inside the rotor iron) links with a much larger amount of localized leakage flux compared to the topmost layer (near the air gap). Consequently, the leakage inductance (L) and leakage reactance ($X = 2\pi fL$) of the bottom layers are significantly higher than those of the top layers.



- **At Starting ($s = 1$):** The rotor is stationary, so the rotor current frequency equals the full supply frequency (e.g., 50 Hz). At this relatively high frequency, the reactance ($2\pi fL$) of the lower portion of the bar becomes so massive that the AC current is forced to avoid the bottom and flow almost entirely in the top portion of the bar (Skin Effect). This crowding drastically reduces the effective cross-sectional area of the conductor, resulting in a very high effective rotor resistance (R_2). Since starting torque $T_{st} \propto R_2$, the motor successfully develops a high starting torque.
- **At Running ($s \approx 0.02$):** The motor is near synchronous speed, making the rotor frequency very low (e.g., 1 to 2.5 Hz). At such low frequencies, the leakage reactance term becomes negligible ($X \approx 0$). The current distributes itself uniformly across the entire depth of the bar, guided only by DC resistance. The full cross-sectional area is utilized, resulting in a low effective rotor resistance, which ensures low copper losses and high running efficiency.

Question 2(a): "A rotating magnetic field of constant amplitude can be produced by supplying a balanced three phase voltage source to a balanced three phase winding" Explain. Find out the speed of the rotating magnetic field produced. How can you change the direction of the rotating magnetic field?

SR Sir's Thought Process

This requires defining the time-displaced sinusoidal currents, space-displaced stator windings, writing their mathematical expressions, applying trigonometric identities, and showing the resultant is a traveling wave $1.5F_m \cos(\omega t - \theta)$. Speed is found by setting the phase to zero and taking $d\theta/dt$. Direction is changed by swapping any two supply phases.

Answer:

Explanation of Production of Rotating Magnetic Field (RMF): Consider a stator containing a balanced 3-phase winding. The axes of phases A, B, and C are displaced from each other by 120° mechanically (or electrically for a 2-pole machine) in space. When fed from a balanced 3-phase supply, the instantaneous currents are displaced in time by 120° :

$$i_A(t) = I_m \cos(\omega t) \quad (13)$$

$$i_B(t) = I_m \cos(\omega t - 120^\circ) \quad (14)$$

$$i_C(t) = I_m \cos(\omega t - 240^\circ) = I_m \cos(\omega t + 120^\circ) \quad (15)$$

Each current produces an alternating Magnetomotive Force (MMF) acting along its own fixed spatial axis. Let θ be the spatial angle measured from the axis of phase A. The spatial MMFs are:

$$F_A(\theta, t) = F_m \cos(\omega t) \cos(\theta) \quad (16)$$

$$F_B(\theta, t) = F_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ) \quad (17)$$

$$F_C(\theta, t) = F_m \cos(\omega t + 120^\circ) \cos(\theta + 120^\circ) \quad (18)$$

where F_m is the peak MMF of a single phase.

The total resultant MMF F_{res} is the sum of the individual phase MMFs. Applying the trigonometric identity $\cos X \cos Y = \frac{1}{2}[\cos(X - Y) + \cos(X + Y)]$:

$$F_A = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta)] \quad (19)$$

$$F_B = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta - 240^\circ)] \quad (20)$$

$$F_C = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta + 240^\circ)] \quad (21)$$

Summing these, the backward-traveling terms $(\omega t + \theta)$ form a balanced 3-phase set of phasors that sum precisely to zero. The forward-traveling terms $(\omega t - \theta)$ add algebraically:

$$F_{res}(\theta, t) = \frac{3}{2} F_m \cos(\omega t - \theta) \quad (22)$$

This mathematically proves the production of an RMF with a **constant amplitude** equal to 1.5 times the peak MMF of a single phase ($1.5F_m$).

Speed of the Rotating Magnetic Field: To find the velocity of the field, track a point of constant phase (e.g., the peak flux where the argument of the cosine is zero):

$$\omega t - \theta = 0 \implies \theta = \omega t \quad (23)$$

Differentiating with respect to time gives the angular velocity in electrical radians per second:

$$\frac{d\theta}{dt} = \omega = 2\pi f \quad (24)$$

In mechanical speed (Revolutions Per Minute, RPM), this translates to the Synchronous Speed (N_s):

$$N_s = \frac{120f}{P} \quad (25)$$

where f is the supply frequency and P is the number of poles.

Changing the Direction of the RMF: The direction of rotation of the magnetic field can be reversed by simply **interchanging any two of the three supply lines** connected to the stator terminals (e.g., swapping Phase B with Phase C). This alters the phase sequence from A-B-C to A-C-B, which reverses the sign in the mathematical derivation and causes the field to traverse backwards.

Question 2(b): Derive an expression for torque developed in an induction motor and draw the torque-speed characteristics. At what slip the torque developed in an induction motor will be maximum?

SR Sir's Thought Process

Same as 2022 Question 2(b). Derive T_e using $P_{ag} = 3(I'_2)^2 R'_2 / s$. Draw the curve. Then to find slip at maximum torque, take $dT_e / ds = 0$ resulting in $s_{mT} \approx R'_2 / X'_2$ (assuming negligible stator resistance for simplicity, or full $s_{mT} = R_2 / \sqrt{R_1^2 + (X_1 + X_2)^2}$).

Answer:

Derivation of Torque Expression: Based on the exact equivalent circuit of an induction motor referred to the stator, the per-phase rotor current I'_2 is:

$$I'_2 = \frac{V_1}{\sqrt{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2}} \quad (26)$$

The gross mechanical power developed by the rotor, plus the rotor copper losses, constitutes the power transferred across the air gap from the stator (P_{ag}).

$$P_{ag} = 3(I'_2)^2 \frac{R'_2}{s} \quad (27)$$

The electromagnetic torque developed (T_e) is the air-gap power divided by the synchronous speed in mechanical radians per second ($\omega_s = \frac{2\pi N_s}{60}$):

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{3}{\omega_s} (I'_2)^2 \frac{R'_2}{s} \quad (28)$$

Substituting the expression for I'_2 , we obtain the definitive equation for developed torque:

$$T_e = \frac{3}{\omega_s} \frac{V_1^2 \frac{R'_2}{s}}{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2} \text{ N-m} \quad (29)$$

Condition for Maximum Torque: To find the slip at which maximum torque is developed (s_{mT}), we apply the Maximum Power Transfer Theorem to the equivalent circuit, or mathematically differentiate T_e with respect to s and equate to zero ($\frac{dT_e}{ds} = 0$). Torque is maximized when the variable equivalent resistance term ($\frac{R'_2}{s}$) becomes equal to the magnitude of the equivalent impedance looking back into the source:

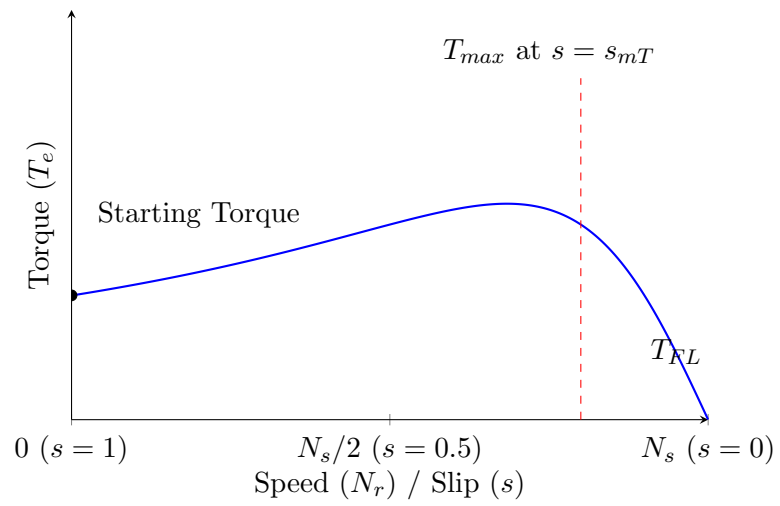
$$\frac{R'_2}{s_{mT}} = \sqrt{R_1^2 + (X_1 + X'_2)^2} \quad (30)$$

Solving for s_{mT} :

$$s_{mT} = \frac{R'_2}{\sqrt{R_1^2 + (X_1 + X'_2)^2}} \quad (31)$$

Note: If stator resistance R_1 is negligibly small, this simplifies to $s_{mT} \approx \frac{R'_2}{X_1 + X'_2}$.

Torque-Speed Characteristics:



Question 3(a): Develop the equivalent circuit of a three phase induction motor and explain how the mechanical power developed is taken care in the equivalent circuit.

SR Sir's Thought Process

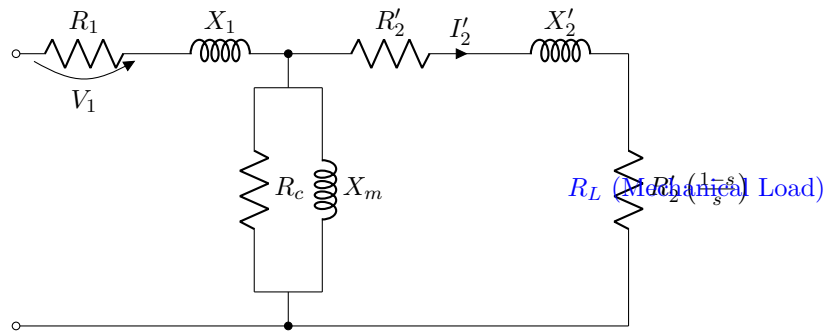
1. Draw the exact equivalent circuit using circuitikz. 2. Explain the physical division of the rotor resistance parameter R_2/s into R_2 (copper loss) and $R_2(1-s)/s$ (mechanical power equivalent).

Answer:

Development of the Equivalent Circuit: The induction motor is essentially a rotating transformer. The stator acts as the primary and the rotor acts as a short-circuited secondary. Let R_1, X_1 be the stator resistance and leakage reactance. Let R_c, X_m be the core loss resistance and magnetizing reactance forming the no-load parallel branch. When the motor runs at slip s , the induced rotor voltage is sE_2 and the rotor reactance is sX_2 . The rotor current is:

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \quad (32)$$

Dividing the numerator and denominator by s mathematically transforms the rotating secondary into an equivalent stationary secondary with fixed voltage E_2 , fixed reactance X_2 , and a variable resistance $\frac{R_2}{s}$. Referring these rotor parameters to the stator side by multiplying with the turns ratio squared yields R'_2 and X'_2 . This results in the complete per-phase equivalent circuit:



How Mechanical Power is Taken Care of: In the equivalent circuit, the total power transferred from stator to rotor (Air Gap Power, P_{ag}) is dissipated entirely in the effective rotor resistance term $\left(\frac{R'_2}{s}\right)$. However, physically, the rotor only dissipates heat through its actual ohmic resistance R'_2 . The rest of the power is converted into mechanical work. To model this electrically, we split the term $\left(\frac{R'_2}{s}\right)$ into two distinct parts:

$$\frac{R'_2}{s} = R'_2 + R'_2 \left(\frac{1-s}{s}\right) \quad (33)$$

- R'_2 (**Actual Resistance**): The power consumed by this component ($I_2'^2 R'_2$) represents the actual thermal $I^2 R$ copper losses in the rotor windings.
- $R'_2 \left(\frac{1-s}{s}\right)$ (**Fictitious Load Resistance R_L**): The power consumed by this term ($I_2'^2 R'_2 \frac{1-s}{s}$) accurately represents the **Gross Mechanical Power Developed (P_d)** by the motor.

Thus, the physical rotating mechanical load on the motor shaft is elegantly replaced by a purely stationary, variable electrical resistor $R_L = R'_2 \left(\frac{1-s}{s}\right)$ in the equivalent circuit.

Question 3(b): Describe different tests to determine the equivalent circuit parameters of a three phase induction motor. How can you calculate the equivalent circuit parameters from these test results?

SR Sir's Thought Process

This is identical to 2022 Question 3(b). Standard 3 tests: DC resistance test, No-Load test, Blocked Rotor test.

Answer:

To determine the parameters $(R_1, X_1, R_c, X_m, R'_2, X'_2)$, three distinct standard tests are performed:

1. DC Test for Stator Resistance (R_1) A low DC voltage is applied across two stator terminals to circulate rated current. Voltage (V_{dc}) and current (I_{dc}) are measured. For a star-connected motor, the resistance per phase is $R_{dc} = \frac{V_{dc}}{2I_{dc}}$. Because AC resistance is higher due to skin effect, it is scaled: $R_1 \approx 1.2 \times R_{dc}$.

2. No-Load Test The motor is run without any mechanical load at rated voltage (V_0) and rated frequency. It spins very close to synchronous speed ($s \approx 0$). Thus, the load resistor $R'_2 \frac{1-s}{s}$ becomes virtually infinite, effectively opening the rotor branch. *Measurements:* V_0 (line voltage), I_0 (no-load line current), P_0 (total 3-phase input power). *Calculations (per phase):* $V_{0ph} = V_0/\sqrt{3}$, $P_{0ph} = P_0/3$. The no-load power factor is $\cos \phi_0 = \frac{P_{0ph}}{V_{0ph}I_0}$. The working current is $I_c = I_0 \cos \phi_0$, and magnetizing current is $I_m = I_0 \sin \phi_0$. The shunt parameters are calculated as:

$$R_c = \frac{V_{0ph}}{I_c} \quad (34)$$

$$X_m = \frac{V_{0ph}}{I_m} \quad (35)$$

3. Blocked Rotor Test The rotor shaft is physically locked so it cannot rotate ($s = 1$). A reduced 3-phase voltage (V_{br}) is applied such that rated full-load current (I_{br}) flows. Because the voltage is very low, the magnetizing branch current is negligible, and all current effectively flows through the series impedances. *Measurements:* V_{br} (line voltage), I_{br} (line current), P_{br} (total 3-phase power). *Calculations (per phase):* $V_{br,ph} = V_{br}/\sqrt{3}$, $P_{br,ph} = P_{br}/3$. Equivalent blocked rotor impedance: $Z_{br} = \frac{V_{br,ph}}{I_{br}}$. Equivalent blocked rotor resistance: $R_{br} = \frac{P_{br,ph}}{I_{br}^2}$. Equivalent blocked rotor reactance: $X_{br} = \sqrt{Z_{br}^2 - R_{br}^2}$.

Extracting Parameters: Since $R_{br} = R_1 + R'_2$, the rotor resistance referred to stator is $R'_2 = R_{br} - R_1$. Since $X_{br} = X_1 + X'_2$, it is generally assumed that $X_1 = X'_2 = \frac{X_{br}}{2}$ for standard motor designs.

Question 4(a): i) If the stator impedance of an induction motor is neglected show that:

$$T_e/T_{em} = 2/(s_{mT}/s + s/s_{mT})$$

ii) From the equivalent circuit of a poly phase induction motor, obtain the following relations:

$$1) I_{2st}/I_2 = \sqrt{[(s^2 + s_{mT}^2)/(s^2(1 + s_{mT}^2))]}$$

$$2) I_{2mT}/I_2 = \sqrt{[0.5(1 + (s_{mT}/s)^2)]}$$

SR Sir's Thought Process

This is identical to 2022 Question 4(a). I will reproduce the algebraic derivations ensuring absolute rigor.

Answer:

i) Proof of Torque Ratio: Neglecting stator impedance ($R_1 \approx 0, X_1 \approx 0$), the basic torque equation is:

$$T_e = \frac{kV_1^2 \frac{R_2}{s}}{\left(\frac{R_2}{s}\right)^2 + X_2^2} = \frac{kV_1^2 s R_2}{R_2^2 + s^2 X_2^2} \quad (36)$$

Maximum torque occurs at slip $s_{mT} = \frac{R_2}{X_2}$, which means $X_2 = \frac{R_2}{s_{mT}}$. Substitute $s = s_{mT}$ into the torque equation to find maximum torque T_{em} :

$$T_{em} = \frac{kV_1^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 X_2^2} = \frac{kV_1^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 \left(\frac{R_2}{s_{mT}}\right)^2} = \frac{kV_1^2 s_{mT}}{2R_2} \quad (37)$$

Dividing T_e by T_{em} :

$$\frac{T_e}{T_{em}} = \left(\frac{kV_1^2 s R_2}{R_2^2 + s^2 X_2^2}\right) \div \left(\frac{kV_1^2 s_{mT}}{2R_2}\right) = \frac{2s R_2^2}{s_{mT}(R_2^2 + s^2 X_2^2)} \quad (38)$$

Divide numerator and denominator by R_2^2 :

$$\frac{T_e}{T_{em}} = \frac{2s}{s_{mT} \left(1 + s^2 \left(\frac{X_2}{R_2}\right)^2\right)} = \frac{2s}{s_{mT} \left(1 + \frac{s^2}{s_{mT}^2}\right)} = \frac{2s}{s_{mT} + \frac{s^2}{s_{mT}}} \quad (39)$$

Finally, divide numerator and denominator by s :

$$\boxed{\frac{T_e}{T_{em}} = \frac{2}{\frac{s_{mT}}{s} + \frac{s}{s_{mT}}}} \quad (\text{Proved}) \quad (40)$$

ii) Proof of Current Relations: The rotor current at any generic slip s is:

$$I_2 = \frac{V_1}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} = \frac{sV_1}{\sqrt{R_2^2 + s^2 X_2^2}} \quad (41)$$

Substitute $R_2 = s_{mT} X_2$:

$$I_2 = \frac{sV_1}{\sqrt{s_{mT}^2 X_2^2 + s^2 X_2^2}} = \frac{sV_1}{X_2 \sqrt{s_{mT}^2 + s^2}} \quad (42)$$

Relation 1: I_{2st}/I_2 At starting, $s = 1$. The starting current I_{2st} is:

$$I_{2st} = \frac{V_1}{X_2 \sqrt{s_{mT}^2 + 1}} \quad (43)$$

Take the ratio:

$$\frac{I_{2st}}{I_2} = \frac{\frac{V_1}{X_2\sqrt{s_{mT}^2+1}}}{\frac{sV_1}{X_2\sqrt{s_{mT}^2+s^2}}} = \frac{\sqrt{s^2+s_{mT}^2}}{s\sqrt{1+s_{mT}^2}} = \sqrt{\frac{s^2+s_{mT}^2}{s^2(1+s_{mT}^2)}} \quad (44)$$

$$\boxed{\frac{I_{2st}}{I_2} = \sqrt{\frac{s^2+s_{mT}^2}{s^2(1+s_{mT}^2)}}} \quad (\text{Proved}) \quad (45)$$

Relation 2: I_{2mT}/I_2 At max torque, $s = s_{mT}$. Current I_{2mT} is:

$$I_{2mT} = \frac{s_{mT}V_1}{X_2\sqrt{s_{mT}^2+s_{mT}^2}} = \frac{V_1}{X_2\sqrt{2}} \quad (46)$$

Take the ratio:

$$\frac{I_{2mT}}{I_2} = \frac{\frac{V_1}{X_2\sqrt{2}}}{\frac{sV_1}{X_2\sqrt{s^2+s_{mT}^2}}} = \sqrt{\frac{s^2+s_{mT}^2}{2s^2}} = \sqrt{0.5 \left(\frac{s^2}{s^2} + \frac{s_{mT}^2}{s^2} \right)} \quad (47)$$

$$\boxed{\frac{I_{2mT}}{I_2} = \sqrt{0.5 \left(1 + \left(\frac{s_{mT}}{s} \right)^2 \right)}} \quad (\text{Proved}) \quad (48)$$

Question 4(b): Why starters are necessary for starting an induction motor? What are the various types of starters used for starting of squirrel cage induction motor? Describe with circuit diagram the working of any one type of starter for starting squirrel cage induction motor and hence derive an expression for starting torque in terms of full-load torque.

SR Sir's Thought Process

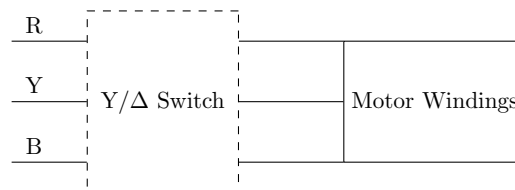
1. Necessity: high starting current (5-7 times rated) due to absence of back EMF at $s = 1$. Can cause massive voltage dips in the local grid and thermal damage. 2. Types for SCIM: DOL, Stator Resistance, Auto-transformer, Star-Delta. 3. Detail Star-Delta as it's the most common/examinable. Draw simple schematic showing switch from Star to Delta. Derive $T_{st} = (1/3)T_{st,DOL}$.

Answer:

Necessity of Starters: At the instant of starting ($s = 1$), an induction motor behaves like a short-circuited transformer. Because the rotor is stationary, it has zero back-EMF, causing it to draw an immense starting current from the supply lines—typically 5 to 7 times its rated full-load current. If connected directly to the grid, this massive current surge can: 1. Cause severe thermal stress and potentially burn out the motor windings. 2. Produce massive voltage drops in the supply lines, causing other equipment connected to the same grid to malfunction or shut down. To mitigate this, starters are used to temporarily reduce the voltage applied to the motor terminals during the starting period, thereby limiting the inrush current.

Types of Starters for Squirrel Cage Induction Motors (SCIM): 1. Direct On-Line (DOL) Starter (for small motors only) 2. Stator Resistance/Reactor Starter 3. Auto-Transformer Starter 4. **Star-Delta ($Y - \Delta$) Starter**

Working of a Star-Delta Starter: The Star-Delta starter is the most widely used technique for medium-sized SCIMs designed to run in Delta configuration. It involves a multi-pole switch that initially connects the motor stator windings in a Star (Y) formation during start-up, and later switches them to a Delta (Δ) formation once the motor attains roughly 80% of its normal speed.



Operation and Derivation: Let V_L be the rated line voltage. If the motor were started directly in Delta (DOL): The phase voltage across windings is $V_{ph} = V_L$. Phase starting current $I_{st,\Delta} = \frac{V_L}{Z_{sc}}$. Line starting current $I_{L,\Delta} = \sqrt{3}I_{st,\Delta} = \sqrt{3}\frac{V_L}{Z_{sc}}$. The starting torque in DOL is: $T_{st,DOL} \propto I_{st,\Delta}^2 \Rightarrow T_{st,DOL} = k \left(\frac{V_L}{Z_{sc}} \right)^2$.

When started using the Star-Delta switch: The switch connects windings in Star. The phase voltage is reduced by $1/\sqrt{3}$, so $V_{ph,Y} = \frac{V_L}{\sqrt{3}}$. Phase starting current $I_{st,Y} = \frac{V_L/\sqrt{3}}{Z_{sc}} = \frac{1}{\sqrt{3}} \left(\frac{V_L}{Z_{sc}} \right)$. Because it is in Star, the line current equals phase current: $I_{L,Y} = I_{st,Y} = \frac{1}{\sqrt{3}} \left(\frac{V_L}{Z_{sc}} \right)$. Comparing the line currents from the grid:

$$I_{L,Y} = \frac{1}{3} I_{L,\Delta} \quad (49)$$

The starter successfully limits the line starting current to exactly one-third of the DOL current.

Starting Torque Expression: Torque is proportional to the square of the phase current passing through the windings ($T \propto I_{ph}^2$).

$$T_{st,Y} \propto I_{st,Y}^2 = k \left(\frac{V_L/\sqrt{3}}{Z_{sc}} \right)^2 = k \frac{1}{3} \left(\frac{V_L}{Z_{sc}} \right)^2 \quad (50)$$

Therefore:

$$\boxed{T_{st,Y-\Delta} = \frac{1}{3} T_{st,DOL}} \quad (51)$$

Using the fundamental relation $\frac{T_{st,DOL}}{T_{fl}} = \left(\frac{I_{st,\Delta}}{I_{fl}} \right)^2 s_{fl}$, the expression for Star-Delta starting torque in terms of full-load torque is:

$$\boxed{\frac{T_{st,Y-\Delta}}{T_{fl}} = \frac{1}{3} \left(\frac{I_{st,\Delta}}{I_{fl}} \right)^2 s_{fl}} \quad (52)$$

Question 5(a): The power input to a 6-pole, 50 Hz, 3-phase induction motor is 700 W at no-load and 10 kW at full-load. The no-load copper losses may be assumed negligible while the full-load stator and rotor copper losses are 295 W and 310 W respectively. Find the full-load speed, shaft torque and efficiency of the motor assuming the rotational and core losses to be equal.

SR Sir's Thought Process

1. Parse no-load data: $P_{in,NL} = 700$ W. Since $I^2 R_{NL} \approx 0$, the 700 W is entirely constant losses ($P_{core} + P_{rot}$). Since they are equal, each is 350 W. 2. Parse full-load data: $P_{in} = 10000$ W. Stator Cu loss (P_{scl}) = 295 W. Rotor Cu loss (P_{rcl}) = 310 W. 3. Air gap power $P_{ag} = P_{in} - P_{scl} - P_{core} = 10000 - 295 - 350 = 9355$ W. 4. Slip $s = P_{rcl}/P_{ag} = 310/9355 = 0.03313$. 5. Speed $N_r = (1 - s)N_s$. $N_s = 1000$ RPM. $N_r = 966.86$ RPM. 6. Mechanical power developed $P_d = P_{ag} - P_{rcl} = 9355 - 310 = 9045$ W. 7. Output shaft power $P_{out} = P_d - P_{rot} = 9045 - 350 = 8695$ W. 8. Torque $T_{sh} = P_{out}/\omega_r$. 9. Efficiency $\eta = P_{out}/P_{in}$.

Answer:

Step 1: Evaluation of Constant Losses At no-load, the power input goes entirely towards supplying the constant losses (since no-load copper losses are stated to be negligible). Total Constant Losses = $P_{core} + P_{rotational} = 700$ W. Given that core losses and rotational losses are equal:

$$P_{core} = P_{rotational} = \frac{700}{2} = 350 \text{ W} \quad (53)$$

Step 2: Full-Load Power Flow Full-load input power $P_{in} = 10 \text{ kW} = 10000$ W. Stator Copper Loss $P_{scl} = 295$ W. The power transferred across the air gap (P_{ag}) is:

$$P_{ag} = P_{in} - P_{scl} - P_{core} = 10000 - 295 - 350 = \mathbf{9355 \text{ W}} \quad (54)$$

The gross mechanical power developed by the rotor (P_d) is the air gap power minus the rotor copper losses ($P_{rcl} = 310$ W):

$$P_d = P_{ag} - P_{rcl} = 9355 - 310 = \mathbf{9045 \text{ W}} \quad (55)$$

The net shaft output power (P_{out}) is the mechanical power minus rotational losses:

$$P_{out} = P_d - P_{rotational} = 9045 - 350 = \mathbf{8695 \text{ W}} = 8.695 \text{ kW} \quad (56)$$

Step 3: Calculating Full-Load Speed The slip s can be found from the power ratio:

$$s = \frac{\text{Rotor Copper Loss}}{P_{ag}} = \frac{310}{9355} = 0.033137 \text{ (or 3.31\%)} \quad (57)$$

Synchronous speed $N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000$ RPM. Full-load rotor speed (N_r):

$$N_r = (1 - s)N_s = (1 - 0.033137) \times 1000 = \mathbf{966.86 \text{ RPM}} \quad (58)$$

Step 4: Calculating Shaft Torque Angular velocity of rotor $\omega_r = \frac{2\pi N_r}{60} = \frac{2\pi \times 966.86}{60} = 101.25$ rad/sec. Shaft Torque (T_{sh}):

$$T_{sh} = \frac{P_{out}}{\omega_r} = \frac{8695}{101.25} = \mathbf{85.87 \text{ N-m}} \quad (59)$$

Step 5: Calculating Efficiency

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% = \frac{8695}{10000} \times 100\% = \mathbf{86.95\%} \quad (60)$$

Question 5(b): The resistance and reactance (equivalent) values of a double-cage induction motor for stator, outer and inner cage are 0.25, 1.0 and 0.15 Ω resistance and 3.5, zero and 3.0 Ω reactance respectively. Find the starting torque if the phase voltage is 250 V and the synchronous speed is 1000 rpm.

SR Sir's Thought Process

1. Organize parameters: Stator: $R_1 = 0.25$, $X_1 = 3.5$. Outer cage: $R_o = 1.0$, $X_o = 0$. Inner cage: $R_i = 0.15$, $X_i = 3.0$.
2. At starting, slip $s = 1$. The two cages are in parallel across the rotor bus.
3. Calculate equivalent parallel impedance $Z_p = (Z_o Z_i) / (Z_o + Z_i)$.
4. Calculate total impedance $Z_{eq} = Z_1 + Z_p$.
5. Calculate starting current $I_{st} = V_{ph} / |Z_{eq}|$.
6. Calculate total equivalent rotor resistance $R_p = \text{Re}\{Z_p\}$.
7. Calculate torque $T_{st} = (3/\omega_s) I_{st}^2 R_p$.

Answer:

Given Data (per phase values): Phase Voltage $V_{ph} = 250$ V Synchronous speed $N_s = 1000$ RPM $\Rightarrow \omega_s = \frac{2\pi \times 1000}{60} = 104.72$ rad/sec Stator impedance $Z_1 = R_1 + jX_1 = 0.25 + j3.5 \Omega$ Outer cage impedance (at $s = 1$) $Z_o = R_o + jX_o = 1.0 + j0 \Omega$ Inner cage impedance (at $s = 1$) $Z_i = R_i + jX_i = 0.15 + j3.0 \Omega$

Step 1: Equivalent Rotor Impedance at Starting At start ($s = 1$), the outer and inner cages are electrically in parallel. The equivalent rotor impedance Z_p is:

$$Z_p = \frac{Z_o \cdot Z_i}{Z_o + Z_i} = \frac{(1.0) \cdot (0.15 + j3.0)}{(1.0) + (0.15 + j3.0)} = \frac{0.15 + j3.0}{1.15 + j3.0} \quad (61)$$

Rationalizing the denominator:

$$Z_p = \frac{(0.15 + j3.0)(1.15 - j3.0)}{(1.15)^2 + (3.0)^2} \quad (62)$$

$$= \frac{0.1725 - j^2(9.0) + j(3.45) - j(0.45)}{1.3225 + 9.0} \quad (63)$$

$$= \frac{9.1725 + j3.0}{10.3225} \quad (64)$$

$$= 0.8886 + j0.2906 \Omega \quad (65)$$

The equivalent rotor resistance $R_p = 0.8886 \Omega$.

Step 2: Total Motor Impedance Total equivalent impedance $Z_{eq} = Z_1 + Z_p$:

$$Z_{eq} = (0.25 + j3.5) + (0.8886 + j0.2906) = 1.1386 + j3.7906 \Omega \quad (66)$$

Magnitude squared of total impedance:

$$|Z_{eq}|^2 = (1.1386)^2 + (3.7906)^2 = 1.296 + 14.368 = 15.664 \Omega^2 \quad (67)$$

Step 3: Stator Phase Current at Starting

$$I_{st}^2 = \frac{V_{ph}^2}{|Z_{eq}|^2} = \frac{(250)^2}{15.664} = \frac{62500}{15.664} = 3989.9 \text{ A}^2 \quad (68)$$

(The actual starting phase current magnitude is $|I_{st}| = \sqrt{3989.9} = 63.17$ A).

Step 4: Starting Torque Calculation The power transferred across the air gap at start ($s = 1$) is the power dissipated in the equivalent rotor resistance R_p .

$$P_{ag,st} = 3I_{st}^2 R_p = 3 \times 3989.9 \times 0.8886 = 10636 \text{ W} \quad (69)$$

The starting torque T_{st} is:

$$T_{st} = \frac{P_{ag,st}}{\omega_s} = \frac{10636}{104.72} = \mathbf{101.57 \text{ N-m}} \quad (70)$$

— **END OF 2023 PART I SOLUTIONS** —
May you master these concepts with absolute clarity.